

Exercise Lecture 8: Black formula

The payoff of the call on a forward is $(F_T - K)^+$.

→ Reformulate as an exchange option as follows:

$$\text{Asset 1: } S_1(t) = P(t, T) F_t$$

$$\text{Asset 2: } S_2(t) = P(t, T) K$$

$$P(T, T) = 1 \Rightarrow S_1(T) = F_T \text{ and } S_2(T) = K$$

So the payoff of this exchange option is

$$(S_1(T) - S_2(T))^+ = (F_T - K)^+.$$

The Margrabe formula is given by

$$\begin{aligned} C_0 &= S_1(0) N(d_1) - S_2(0) N(d_2) \\ &= P(0, T) (F_0 N(d_1) - K N(d_2)) \end{aligned}$$

$$\text{with } d_1 = \frac{\ln\left(\frac{S_1}{S_2}\right) + \frac{1}{2}\Sigma^2 T}{\Sigma\sqrt{T}} = \frac{\ln(F_0/K) + \frac{1}{2}\Sigma^2 T}{\Sigma\sqrt{T}}$$

$$\text{and } d_2 = \frac{\ln\left(\frac{S_1}{S_2}\right) - \frac{1}{2}\Sigma^2 T}{\Sigma\sqrt{T}} = \frac{\ln(F_0/K) - \frac{1}{2}\Sigma^2 T}{\Sigma\sqrt{T}}$$

where Σ is the volatility of $\frac{S_1}{S_2}$.

$$\frac{S_1}{S_2} = \frac{F_0}{K} \rightarrow \text{volatility is } 0$$